
Addendum for *MISO DPP 2016 August West Area Phase 3 Study*

February 5, 2020

This addendum contains updated study results for the report entitled MISO DPP 2016 August West Area Phase 3 Study and dated March 14, 2019. Results have been updated to reflect Network Upgrade (NU) changes after the withdrawal of generation project J598 (300 MW). The withdrawal of this project does not trigger a restudy for remaining DPP-2016-August West area projects.

1.1 Withdrawal of J598

Withdrawal of generation project J598 was studied to determine continued need for network upgrades and calculate update cost allocation percentages. After the withdrawal of J598, the following Network Upgrades (NUs) are determined to still be needed:

Table 1: Thermal Network Upgrades Needed in MISO Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
J530 POI-Montezuma 345 kV	1010.0	MEC	1118.8	110	CEII	P1	J541,J555
J530 POI-Montezuma 345 kV	1010.0	MEC	1158.6	114	CEII	P2-P7	J541,J555
J530 POI-Hills 345 kV	956.0	MEC	1061.3	110	CEII	P0	J476,J541,J555,J583,J611
J530 POI-Hills 345 kV	1152.0	MEC	1338.3	117	CEII	P1	J541,J555
J530 POI-Hills 345 kV	1152.0	MEC	1518.9	121	CEII	P2-P7	J541,J555

Table 2: Voltage Network Upgrades Needed in MISO Analysis

Network Upgrade	Cost	NUs Type
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	Reactive Power NU

The upgrades that are still required even with the withdrawal of J598 are to be reallocated to the remaining projects by the percentages documented in the table below.

Table 3: Network Upgrades ERIS Cost Allocation

Constraint	J302	J476	J503	J512	J541	J555	J569	J583	J587	J590	J598	J611	J614
75730 J530 POI 345 635730 MNTZUMA3 345 1	0.00%	0.00%	0.00%	0.00%	67.85%	32.15%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
75730 J530 POI 345 636400 HILLS 3 345 1	0.00%	8.11%	0.00%	0.00%	53.75%	25.55%	0.00%	9.12%	0.00%	0.00%	0.00%	3.47%	0.00%
2x150 Mvar switched cap bank at Hills 345 kV (636400)	1.21%	8.35%	1.18%	3.87%	46.42%	19.76%	1.73%	9.73%	2.52%	1.69%	0.00%	3.55%	0.00%

The withdrawal of J598 did result in the determination that certain Network Upgrades were no longer required for the 2016 August West cycle.

Table 4: NRIS Upgrades Determined to be no longer needed

Network Upgrade	Cost	NUs Type
Montezuma-J530 POI 345 kV (NRIS)	\$100,000	NRIS

Table 5: Ameren LPC Upgrades Determined to be no longer needed

Constraint	Mitigation	Cost (\$)
Zackary 345/161 kV transformer	Add Second 560 MVA 345/161 kV transformer	\$7,000,000
Adair-Zackary 161 kV	Add second 161 kV line between Adair and Zachary	\$2,000,000

Table 6: Affected System Network Upgrades Determined to be no longer needed

Constraint	Mitigation	Cost (\$)
Novelty 161/69 kV Transformer	Add Second 56 MVA Novelty 161/69 kV Transformer	\$3,280,000
South River-Emerson 161	Upgrade South River Disconnect Switches	\$100,000

1.2 Conclusion

A restudy is determined to be not needed for the remaining DPP-2016-August West area projects. The updated cost allocation for the ERIS upgrades that are still required were calculated. Network Upgrades that are no longer required were also determined.